

Computing anharmonic infrared spectra of polycyclic aromatic hydrocarbons using machine learning molecular dynamics

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ABSTRACT

We introduce a machine learning molecular dynamics (MLMD) approach to calculate the anharmonic infrared (IR) absorption spectra of polycyclic aromatic hydrocarbons (PAHs), key carriers of interstellar aromatic IR bands. This method accounts for temperature effects in a molecule-specific way and achieves accuracy comparable to conventional quantum chemical calculations at a fraction of the cost, scaling linearly with system size. We applied MLMD to calculate the anharmonic spectra of 1704 PAHs in the NASA Ames PAH IR Spectroscopic Data base with up to 216 carbon atoms at different temperatures, demonstrating its capability for high-throughput spectral calculations of large molecular systems. Our results highlight MLMD's potential to enable the development of extensive molecular spectral data sets, enhancing data-driven analyses of astronomical IR spectra, particularly in anticipation of upcoming observations from the *James Webb Space Telescope*.

Key words: astronomical data bases: miscellaneous – software: data analysis – ISM: molecules – infrared: ISM.

1 INTRODUCTION

In the mid-1970s, distinct infrared (IR) signals were identified in the universe, notably at wavelengths of 3.3, 6.2, 7.7, 8.6, 11.2, and 12.7 μm . These signals were later linked to the vibrations of C–H and C–C bonds in aromatic molecules, known as aromatic IR bands (AIB; Leger & Puget 1984; Allamandola, Tielens & Barker 1985). Such emissions are primarily attributed to polycyclic aromatic hydrocarbons (PAHs) after the absorption of ultraviolet (UV) photons. PAHs are recognized as key contributors to the evolution of the interstellar medium, influencing processes such as gas heating, ionization balance, and star formation (Tielens 2008). Determining the population, composition, size distribution, charge state, and chemical structures of interstellar PAHs provides valuable insight into the physical conditions of their host galaxies. However, this remains a significant challenge due to the structural diversity of PAHs. Variations in carbon-ring arrangements, side groups, substitutions, charge states, and isotopologues lead to distinct IR emission characteristics, each governed by specific vibration modes (Peeters et al. 2021).

In order to tackle these challenges, quantum chemical calculations (QCCs) are now an indispensable aid in AIB analysis. Using QCCs, publicly available spectral data bases have been established for AIB analysis, such as the NASA Ames PAH IR Spectroscopic Data base (PAHdb) (Boersma et al. 2014; Bauschlicher Jr et al. 2018; Mattioda et al. 2020). The availability of these data bases

enables extensive data-driven analysis of AIBs, providing a more robust interpretation of their features (see, e.g. Cami 2011; Boersma, Bregman & Allamandola 2013, 2015; Sadjadi, Zhang & Kwok 2015; Shannon & Boersma 2019; Maragkoudakis, Peeters & Ricca 2020; Ricca, Boersma & Peeters 2021). Furthermore, these data bases have facilitated the analysis of astrochemical components using machine learning (ML) models, demonstrating significant computational efficiency (Kovács et al. 2020) and helping to uncover relationships between molecular structures and spectra (Meng et al. 2021, 2023). Despite these advancements, current theoretical spectral data bases face three primary limitations:

(i) Harmonic oscillator approximations: Most spectra neglect anharmonic vibrations, leading to significant discrepancies with experimental observations (Mackie et al. 2015, 2018a).

(ii) Neglect of thermal effects: Most theoretical spectra in data bases such as PAHdb are calculated at the ground state (0 K), without incorporating temperature effects. At high temperatures, thermal population of excited vibrational states leads to frequency shifts, intensity redistribution, and the appearance of combination and overtone bands. These effects, coupled with anharmonicity, are not captured by 0 K spectra. Consequently, emission features derived from these spectra involve significant approximations.

(iii) Underrepresentation of large PAHs: Although PAHs with 50 or more C atoms are considered more likely AIB carriers (Sellgren 1984; Allamandola, Tielens & Barker 1989), they are significantly underrepresented in current data bases. For instance, only 444 out of 3020 neutral PAHs in PAHdb (v3.2) exceed this size.

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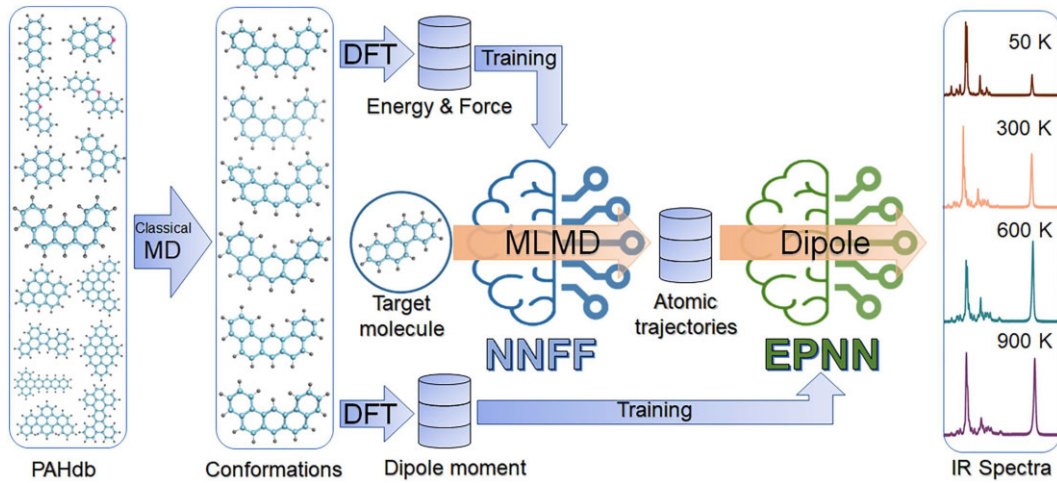


Figure 1. Schematic representation of the workflow for computing anharmonic IR absorption spectra.

These challenges arise mainly from the high computational cost of QCCs. In particular, the computation time for anharmonic molecular spectra using second-order vibrational perturbation theory (VPT2) is prohibitively long, typically only accommodating molecules with $N_C < 25$, where N_C represents the number of carbon atoms in the system (Mackie et al. 2016; Maltseva et al. 2016, 2018; Lemmens et al. 2019; Esposito et al. 2024a). The cost of incorporating temperature effects into these calculations using the Wang–Landau random walk algorithm is even more prohibitive, being orders of magnitude higher due to the need for extensive sampling (Chen 2018; Chen, Luo & Li 2019). Facing the vast diversity of PAH species, these limitations significantly hinder the accurate interpretation of observational data, posing substantial challenges in the era of advanced observational facilities such as the *James Webb Space Telescope (JWST)* (Boersma et al. 2023; Ricca et al. 2024).

As a result, there is an urgent need for a cost-effective and accurate method that incorporates both anharmonic and temperature effects in the construction of spectral data bases to improve our understanding of the origin of AIBs. In this work, we propose a ML-based molecular dynamics (MLMD) approach for computing PAH IR absorption spectra. This method replaces computationally expensive electronic structure calculations with far more efficient ML-based ones. More importantly, it explicitly accounts for anharmonic vibrations and temperature effects. We show that the MLMD approach can reproduce experimental spectra with accuracy comparable to state-of-the-art VPT2 QCCs, while significantly reducing computational time, achieving a scaling approximately linear in N_C .

2 METHODS

Molecular dynamics (MD) is a powerful computational method that simulates the time-dependent behaviour of atomic and molecular systems at a specified temperature by integrating Newton’s equations of motion. This technique uses a potential energy surface (PES) to characterize interatomic interactions and energy changes within the system. When calculating the vibrational spectra of molecules, MD presents significant advantages by explicitly accounting for anharmonic effects, such as band combination, overtones, and mode coupling. This comes at the cost of not capturing some quantum effects whenever normal modes are likely to be found near their ground state. Beyond this fundamental consideration, classical MD has historically faced technical challenges related to accuracy and

transferability. The empirical force fields used to construct the PES are often parametrized for specific molecules, which may hinder their performance when applied to others. Furthermore, although polarizable models exist (Bedrov et al. 2019), classical MD generally does not account for the distribution of electrons and, therefore, cannot inherently deal with dipole moments. To address these challenges, *ab initio* molecular dynamics (AIMD) can be used, where the motion of nuclei is described classically but the force contributions from the electrons are calculated quantum mechanically. However, like the QCC-based VPT2 method, AIMD is computationally expensive, imposing significant constraints on the maximum size of the systems studied.

A solution is to substitute the majority of electronic structure calculations in AIMD with more cost-effective ML computations. Using data points derived from QCCs, ML models can be trained to construct the PES and predict the charge distribution across various molecular configurations. This MLMD methodology has been proven to be highly efficient in previous studies focusing on vibration of diverse molecular types (Gastegger, Behler & Marquetand 2017; Zhou et al. 2021; Du et al. 2024; Schmiedmayer & Kresse 2024; Xu et al. 2024).

In this study, we build two distinct ML models: a neural-network force field (NNFF) for the construction of PES, and an electron-passing neural network (EPNN) model to predict the molecular dipole moment $\mathbf{p}$. As depicted in Fig. 1, our workflow for computing the IR spectra consists of two main phases. The first is a training phase (blue arrows) that includes:

- (i) Classical MD simulations of PAHs to generate conformations;
- (ii) QCCs to obtain their energies, forces, and dipole moments;
- (iii) Training of the NNFF and EPNN models using these data.

The second phase is prediction (orange arrows), including:

- (i) MLMD to obtain conformations of vibrating PAHs via NNFF;
- (ii) Prediction of $\mathbf{p}$ for every conformation using EPNN;
- (iii) Spectrum computation by evaluating the time evolution of $\mathbf{p}$.

The predicted $\mathbf{p}$ is not used as an input for the calculation of energies and forces and therefore does not determine the trajectories, since the interatomic interactions determined by the charge density distribution are already accounted for by the NNFF. A detailed description of the components is provided in the next subsection.

2.1 Training data

The chemical structures of 687 neutral PAH molecules were extracted from the theoretical data set of PAHdb (v3.2; Mattioda et al. 2020) to construct the training data sets. The selected PAHs encompass a diverse array of chemical structures that are of astronomical significance. Classical MD simulations were performed using a custom code to generate random configurations of the atoms in each molecule at 300 K, using the adaptive interatomic reactive empirical bond order potential (Stuart, Tutein & Harrison 2000; Qi et al. 2018; Hanine et al. 2020; Meng & Wang 2023). In these simulations, the molecules reached thermal equilibrium in 200 ps using a canonical Nosé-Hoover thermostat, with a time-step of 0.5 fs. Ten extended structures were extracted from the atomic trajectories of each molecule. As these geometries do not necessarily fall near equilibrium configurations, random Gaussian perturbations were applied to the ground-state atomic coordinates to create 15 more configurations for each molecule to improve the generalizability of the model. Finally, in total, 17 175 configurations (also called molecular conformations) were generated. Our training data set includes 350 molecular conformations of superhydrogenated PAHs, out of a total of 17 175 conformations.

Subsequently, single-point electronic structure QCCs were performed for each of these 17 175 configurations to calculate the potential energy, the forces in the nuclei, and the dipole $\mathbf{p}$, within the framework of density functional theory (DFT) at the B3LYP/4-31G level, as implemented in the GAUSSIAN 16 software package (Frisch et al. 2016). This level of theory offers a favourable balance between accuracy and computational efficiency for large PAHs (Stephens et al. 1994), given its extensive application in PAH IR spectrum studies (Bauschlicher Jr & Bakes 2000; Ricca et al. 2012). We note that the basis set does not include polarization functions, which may introduce more uncertainties than larger basis functions (e.g. 6-31g*) for PAHs containing nitrogen (Ricca et al. 2021, 2024). However, this choice was made to ensure computational efficiency, which is crucial for large-scale sampling. Through these calculations, we generated two data sets for training the NNFF and EPNN models, respectively. The first data set comprises 17 175 energy and 2274 825 force data points, while the second data set includes 17 175 molecular dipole moments.

2.2 NNFF

The NNFF used in this study is based on NeuralIL (Montes-Campos et al. 2021; Carrete et al. 2023; Pan et al. 2024), a refinement of the descriptor-based template introduced by Behler (Behler & Parrinello 2007). NeuralIL employs spherical Bessel descriptors (Kocer, Mason & Erturk 2020) to encode the atomic configurations in a translationally and rotationally invariant fashion, as illustrated in Fig. 2a. We set the cut-off radius of the descriptors to 3.8 Å, with a maximum radial order of 6, ensuring effective capture of local geometric information and sufficient resolution to describe atomic interactions within the molecule.

NNFF utilizes a deep residual network architecture (ResNet; He et al. 2016a), implemented within the JAX framework (Bradbury et al. 2018). A ResNet employs skip connections to learn the residual function between the input and output layers, effectively addressing the data degradation problem that arises with increased network depth (He et al. 2016b). This feature is particularly beneficial for modelling complex molecules such as large PAHs, as it enables the capture of deep dependencies and interaction patterns between atoms through multilevel learning, which ultimately leads to more accurate

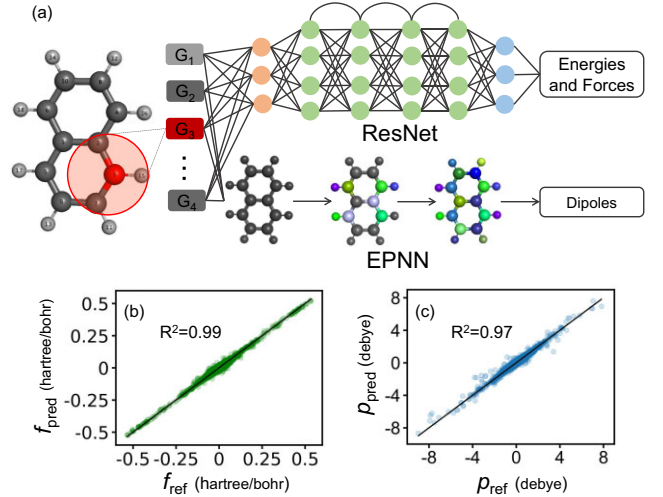


Figure 2. (a) Schematic of the operating principles of each kind of neural network (NNFF and EPNN) during inference. (b) NNFF-predicted atomic forces and (c) EPNN-predicted dipole moments versus the DFT-calculated reference values for 3583 tested PAH configurations.

Table 1. Average MAE of the predicted energy and force from the NNFF, and of the predicted molecular dipole moments from the EPNN, for samples in the training, validation, and test data sets.

	Train	Validation	Test
Energy (hartree per atom)	0.320	0.301	0.319
Force (hartree/bohr)	0.00117	0.00119	0.00115
Dipole (debye)	0.0499	0.0591	0.0557

predictions of physicochemical properties (Xue et al. 2024; Zhang, Gong & Jiang 2024).

In this study, the NNFF is trained based on the previously mentioned DFT-calculated data set of energies E and atomic forces $\mathbf{f}$, with the loss function L defined as a weighted sum of the differences between the predicted and reference values:

$$L = 0.99 \sum \left\langle \frac{0.2}{N_{\text{atom}}} \sum \ln \left[\cosh \left(\frac{\|\mathbf{f}_{\text{pred}} - \mathbf{f}_{\text{ref}}\|_2}{0.2} \right) \right] \right\rangle + 0.01 \sum \left\langle 0.02 \ln \left[\cosh \left(\frac{E_{\text{pred}} - E_{\text{ref}}}{N_{\text{atom}} \times 0.02} \right) \right] \right\rangle, \quad (1)$$

where N_{mol} denotes the total number of molecules, and N_{atom} represents the total number of atoms in a molecule. The subscripts pred and ref denote the values predicted by NNFF and DFT, respectively. $\|\mathbf{f}_{\text{pred}} - \mathbf{f}_{\text{ref}}\|_2$ represents the Euclidean norm of the vector difference between the predicted force and the reference force. The model adopts a core-width sequence of 64:32:16:16 and employs the fully non-linear VELO optimization technique (Metz et al. 2022) for 600 epochs of training to enhance convergence (Carrete et al. 2023).

To evaluate the performance of the model, we conducted tests on a set of configurations for 3583 PAHs that were excluded from the training set. The results presented in Fig. 2b demonstrate that the NNFF model accurately reproduces the forces calculated by DFT, achieving a coefficient of determination (R^2) of approximately 0.99 and a mean absolute error (MAE) of 0.00119 hartree/bohr (see Table 1), indicating strong generalizability to unseen samples. This capability allows the NNFF model to effectively replace DFT

electronic structure calculations in AIMD, providing robust support for subsequent IR spectrum calculations.

The training data of NNFF were computed at the B3LYP/4-31G level of DFT, which is known to systematically overestimate the bond strength. To correct for this, we applied a scaling factor of 0.9578 to the MLMD-predicted vibrational frequencies, following the approach proposed by Bauschlicher & Langhoff (1997).

2.3 EPNN

Predicting the molecular dipole moment $\mathbf{p}$ in a picture of discrete atoms presents a significant challenge due to its dependence on the distribution of atomic partial charges (PC), for which no unique determination method exists. Various PC partitioning schemes can produce different $\mathbf{p}$ for the same molecule. To avoid this issue, we directly train neural networks to predict $\mathbf{p}$, bypassing the need to explicitly calculate PCs as training data, following the approach of Gastegger et al. (2017). However, we utilize a different neural network architecture for enhanced predictive ability. Specifically, we adopt a message-passing neural network known as the EPNN (Metcalf et al. 2021) that guarantees the conservation of charge. In order to achieve charge conservation, the EPNN adopts a graph neural network architecture and ensures antisymmetry of the updates with respect to permutation of the input indices. In the present implementation of the EPNN the node states are extended by concatenation with the spherical Bessel descriptor for each individual atom. Charge conservation is preserved as these descriptors are not modified in the update phase.

The EPNN was trained over 400 epochs using VELO, employing the aforementioned DFT-calculated $\mathbf{p}$ data set. The loss function used for training is defined as

$$L = \sum_{\alpha=1}^3 \sum_{i=1}^{N_{\text{mol}}} \ln \left[\cosh \left(\frac{p_{i,\text{pred}}^{(\alpha)} - p_{i,\text{ref}}^{(\alpha)}}{N_{\text{atom}}} \right) \right], \quad (2)$$

where $p_{i,\text{pred}}^{(\alpha)}$ and $p_{i,\text{ref}}^{(\alpha)}$ represent the predicted and reference values of the α th Cartesian component of $\mathbf{p}$ for the i th molecule. To evaluate the performance of the model, we predicted the $\mathbf{p}_{\text{ref}}$ of 3583 PAH conformations excluded from the training set. As shown in Fig. 2c, there is strong agreement between the predicted and reference values, with a R^2 reaching 0.97 and a MAE of 0.0591 debye, as shown in Table 1.

2.4 Anharmonic IR spectrum

To compute the anharmonic IR absorption spectrum of a PAH, MD simulations were performed to simulate its finite-temperature dynamics based on atomic forces predicted by the trained NNFF. We used the MD implementation in the Atomic Simulation Environment (ASE) (Larsen et al. 2017). Initially, the molecule reaches thermal equilibrium in 200 ps in a canonical ensemble (NVT) at a target temperature controlled by the Nosé–Hoover thermostat, with a time-step of 0.5 fs. The system was then further simulated in a microcanonical ensemble (NVE) for 200 ps, recording atomic configurations at every 1.0 fs during vibration.

These configurations serve as input to the EPNN, which is used to compute $\mathbf{p}(t)$ at each time-step of the vibration. The dipole time autocorrelation function $\langle \mathbf{p}(0) \cdot \mathbf{p}(t) \rangle$, characterizing the time evolution of $\mathbf{p}$, is then subjected to a Fourier transform to obtain the absorption cross-section σ (in electrostatic unit) (Berens & Wilson

1981):

$$\sigma(\omega) = \frac{2\pi\omega(1 - e^{-\hbar\omega/k_B T})D(\omega)}{3\hbar c} \int_0^\infty e^{-i\omega t} \langle \mathbf{p}(0) \cdot \mathbf{p}(t) \rangle dt, \quad (3)$$

where ω denotes the angular frequency, k_B represents the Boltzmann constant, $\hbar$ is the reduced Planck constant, c is the speed of light, T indicates the temperature, t is time, and $D(\omega)$ is the quantum correction factor, which is set to 1.0 in this work. Further explanation of this equation is provided in the [Supplementary Information](#).

σ is connected to the laboratory-measured IR intensity I as:

$$\int_\omega \sigma(\omega) d\omega = 2\pi \cdot 10^5 \frac{c \cdot I(\omega)}{N_A}, \quad (4)$$

where N_A is Avogadro's number. To compare I with the MLMD computed σ , we applied Lorentzian broadening to the experimental spectrum with a full width at half-maximum (FWHM) of 2 cm^{-1} . While the error values vary with the FWHM choice (see the [Supplementary Information](#)), our key conclusion remains unaffected by this selection.

2.5 Reference methods for comparison

To evaluate the accuracy of our model, we compared it with two established QCC methods to calculate the IR spectra of PAHs. The first method, designated as *DFT harmonic*, is a widely adopted hybrid DFT approach for harmonic spectrum calculations. In this method, the vibrational frequencies (normal modes) are calculated from the second derivatives of the potential energy with respect to the positions of the nuclei at the stationary point of the system. IR intensities are determined using the double harmonic approximation, which involves computing the derivatives of $\mathbf{p}$ with respect to the normal modes (Langhoff 1996). The calculations were carried out at the B3LYP/4-31G level of theory using GAUSSIAN 16, ensuring consistency with the QCCs employed in our training data set. It is important to note that the assumption of a harmonic potential frequently underestimates fundamental frequencies, often necessitating empirical scaling factors to align the computational results with the experimental data (Bauschlicher & Langhoff 1997). Despite the transferability issues introduced by these empirical scaling factors, this method remains the most commonly utilized method for molecular IR spectrum calculations due to its relatively low computational cost. For the sake of a fair comparison, we do not apply any scaling factors to the DFT harmonic results, as these factors should ideally be derived from fitting to experimental spectra. The DFT harmonic method was used to compute the harmonic spectra for 49 PAH molecules, for which the experimentally measured spectra are available in the PAHdb.

The second method used for comparison is the VPT2 anharmonic spectrum calculations combined with B3LYP/6-311 + g* DFT, referred to here as *DFT anharmonic*. This approach represents the state-of-the-art for computing the molecular anharmonic IR spectra, and has been implemented in GAUSSIAN 16. It requires the computation of the second, third, and fourth derivatives of the potential, which results in significant computational demands (Nielsen 1951; Mackie et al. 2015). Due to this cost, we applied VPT2 to calculate the anharmonic spectra of 26 PAHs, with the maximum N_C being 32, from the 49 PAHs used for the DFT harmonic method.

The 49 PAHs for which the experimental spectra are available in PAHdb (v3.2), ranging from 10 to 50 carbon atoms, exhibit a variety of chemical structures, as detailed in the [Supplementary Information](#). The experimental IR spectra were obtained using matrix isolation techniques at low temperatures (Hudgins & Allamandola 1995,

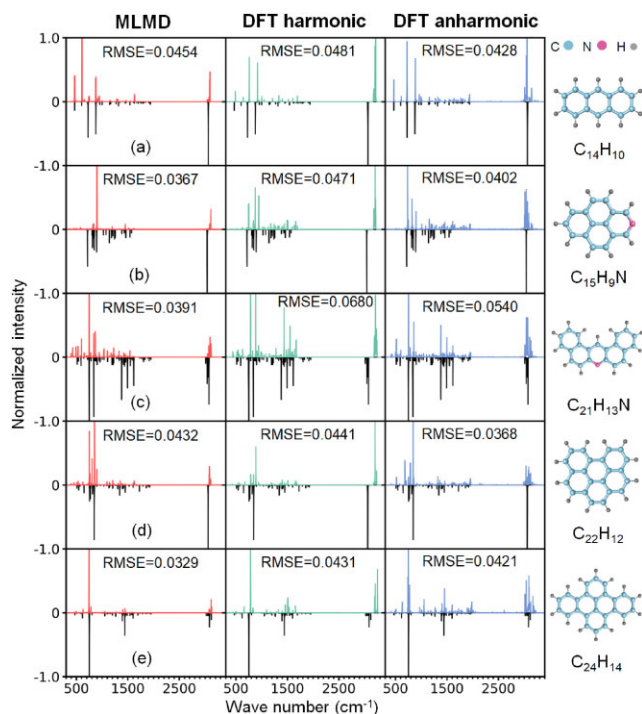


Figure 3. Comparison of IR spectra for five PAHs computed using the MLMD at 50 K (left panel), DFT harmonic (middle panel), and DFT anharmonic (right panel) methods, alongside the experimentally measured spectra. The intensities are normalized to the maximum value in each spectrum. In each panel, the positive curves represent the computed results, while the negative curves correspond to the experimental spectra.

1999; Hudgins & Sandford 1998a, b, c; Mattioda et al. 2003, 2005, 2014, 2017). To facilitate comparison with experimental data, we employed theoretical spectra computed using MLMD at an effective temperature of 50 K, which accounts for molecular vibrations under experimental conditions (Esposito et al. 2024b). Stochastic uncertainty analysis confirms size-independent prediction stability across PAHs, and the model inherently supports non-planar geometries (see [Supplementary Information](#)).

3 RESULTS AND DISCUSSION

To assess the precision of our predictions relative to the experimental data, Fig. 3 presents the computed spectra for five PAHs of increasing sizes, ordered from top to bottom. The negative curves represent the experimental spectra, while the positive curves in the left, middle, and right panels show the spectra computed using MLMD, DFT harmonic, and DFT anharmonic methods, respectively. As seen in the figure, the anharmonic spectra (shown in the left and right panels) exhibit better agreement with the experimental data than the harmonic spectra (middle panel), both in terms of frequency and intensity.

Among the anharmonic spectra, we find that the traditional DFT anharmonic method performs better than the MLMD method for the smallest PAH. However, for larger PAHs, the MLMD method provides more accurate predictions. Specifically, for the molecule $C_{14}H_{10}$, the spectra predicted by DFT anharmonic achieved a root mean square error (RMSE) value of 0.0428. In contrast, the RMSE values for MLMD were higher, at 0.0454, while the DFT harmonic method produced even larger errors of 0.0481, owing to a significant mismatch in band positions, as shown in the middle panels of Fig. 3a.

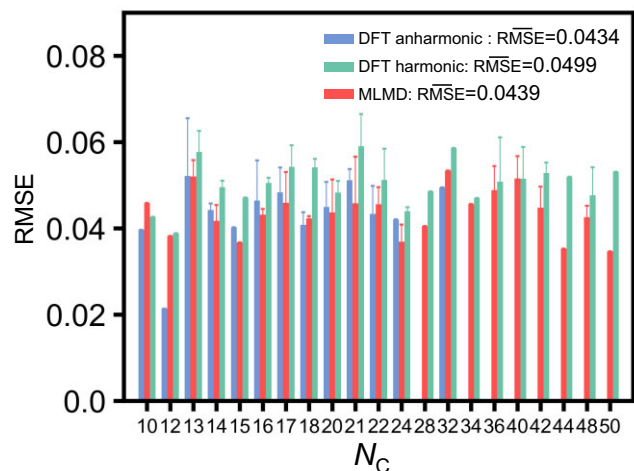


Figure 4. Distribution of RMSE values for spectra predicted by the three computational methods, compared to the experimental spectra of the 49 PAHs. N_C represents the number of carbon atoms in each molecule. Note that the anharmonic DFT calculation is not performed for large PAHs (Section 2.5).

Table 2. Mean RMSE values of IR spectra predicted by MLMD and DFT anharmonic methods, with respect to experimental spectra of the 26 PAHs across spectral regions of key vibrational modes.

Wavelength (μm)	Vibration mode	DFT anharmonic	MLMD
2.6–3.5	CH stretching	0.0534	0.0403
5.2–5.8	Combination	0.0103	0.0044
6–10	CH IP bending	0.0301	0.0273
10–15	CH OOP bending	0.0892	0.1067

For larger PAHs, such as $C_{21}H_{13}N$ and $C_{24}H_{14}$, the MLMD method performs comparably to DFT anharmonic in predicting IR intensities.

Fig. 4 presents an overview of the error distribution in predicting the IR spectra of the 49 PAHs relative to the experimental data, for each of the three computational methods. The results clearly demonstrate that the harmonic spectra exhibit significantly poorer agreement with the experimental data compared to the anharmonic spectra, consistent with the findings of previous studies (Mackie et al. 2015; Maltseva et al. 2016; Lemmens et al. 2019). The precision of the MLMD and DFT anharmonic methods is comparable, with mean RMSE values of 0.0439 and 0.0434, respectively. For PAHs with fewer than 12 carbon atoms, the DFT anharmonic method generally outperforms MLMD. However, as the size of the PAHs increases, MLMD tends to provide comparable predictions than DFT anharmonic in terms of normalized spectra.

To further clarify the differences between MLMD and DFT in specific vibrational regions, a mode-by-mode error analysis is presented in Table 2. The results show that MLMD achieves higher accuracy in the 2.6–3.5, 5.2–5.8, and 6.0–10.0 μm regions, corresponding to C–H stretching, combination bands, and C–H in-plane (IP) bending modes, respectively. In contrast, for out-of-plane (OOP) bending modes in the 10.0–15.0 μm region, the DFT-based anharmonic method demonstrates better agreement with experimental data.

Despite having comparable accuracy, the computational efficiency of MLMD is significantly superior to that of the traditional DFT anharmonic method. Fig. 5 shows the average CPU time, t_{cpu} , required to compute an anharmonic IR spectrum as a function of the molecular size. It is evident that the t_{cpu} for MLMD is initially approximately five times shorter than that for the DFT anharmonic

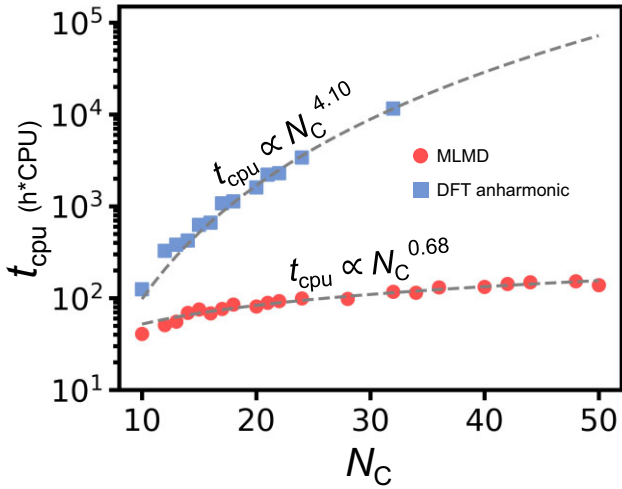


Figure 5. Average CPU time (run time $\times$ number of cores, on a logarithmic scale) required to compute the anharmonic IR spectrum of a PAH as a function of the number of carbon atoms, using the MLMD (solid circles) and DFT anharmonic (solid squares) methods. The DFT anharmonic computations were parallelized on a server with 40 cores (Intel Xeon E5-2680 CPU at 2.40 GHz), while MLMD jobs were run sequentially on a CPU of the same model.

for the smallest PAHs. As the molecular size increases, the difference in t_{cpu} grows, eventually reaching nearly two orders of magnitude for the largest PAH in the set of 49 molecules.

The trend of increasing CPU time with molecular size for MLMD is remarkably different from that of the DFT anharmonic method. To provide a quantitative illustration, a simple power-law fit of type $y = ax^b$ was applied to the data points in Fig. 5, as indicated by the dashed curves. These best-fitting curves show that the DFT computation of anharmonic spectra for large PAHs becomes extremely time consuming, with a computational scaling of $\sim N_C^4$ for the DFT anharmonic method. In contrast, MLMD demonstrates a significantly more efficient scaling of approximately $N_C^{0.7}$. This results in a computational time difference of approximately three orders of magnitude for a PAH with $N_C = 50$. This enhanced efficiency makes MLMD well suited for high-throughput computation of molecular anharmonic spectra, particularly for large PAHs, which are considered key sources of AIBs, and for comprehensively studying the spectroscopic properties of PAHs with extensive structural diversity.

The AIB features are believed to originate from the IR emission of PAHs excited by UV photons. Upon absorbing such a photon, a PAH becomes highly energized, with its effective temperature potentially rising to several hundred or even over a thousand kelvin. Consequently, the low-temperature spectra of PAHs mentioned earlier have limited astronomical relevance in this context. Traditionally, the Wang and Landau method has been used to incorporate temperature effects into a DFT-computed IR spectrum (Wang & Landau 2001; Basire et al. 2009). However, as discussed above, such DFT calculations are computationally expensive. Moreover, the Wang–Landau algorithm often requires extensive sampling to ensure convergence, making it suitable mainly for studying individual, very small PAHs (Chen 2018; Chen et al. 2018; Mackie et al. 2018b, 2021; Chakraborty et al. 2021). In contrast, the MLMD approach employed in this work explicitly incorporates temperature effects, providing a more efficient solution due to its high computational efficiency and accuracy.

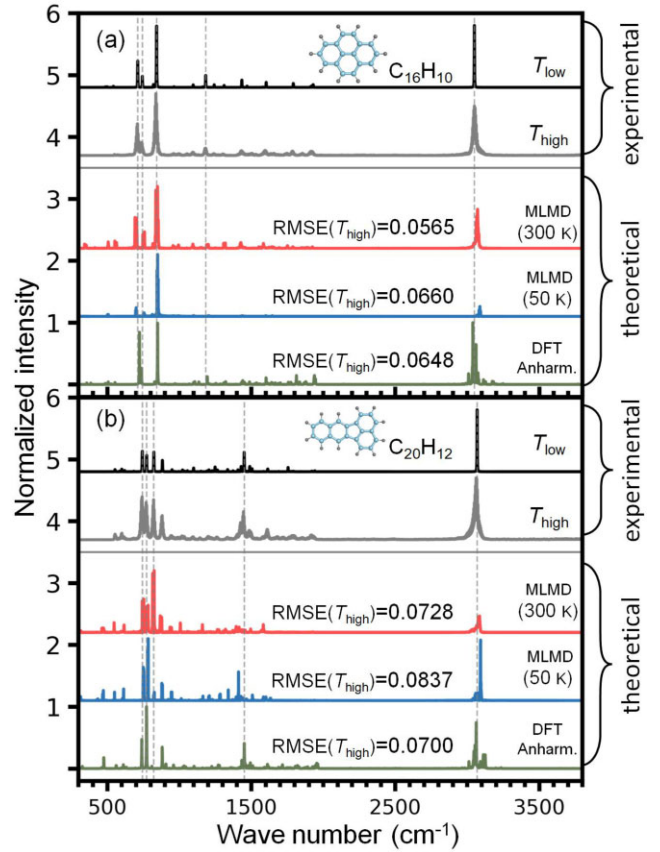


Figure 6. Comparison of IR spectra for pyrene $\text{C}_{16}\text{H}_{10}$ (a) and benzo[k]fluoranthene $\text{C}_{20}\text{H}_{12}$ (b) between experimental data (top panels) and theoretical predictions by MLMD at two different temperatures and by the DFT anharmonic method (bottom panels).

To demonstrate the precision, we compared theoretical anharmonic spectra with experimental IR spectra of two PAHs, pyrene and benzo[k]fluoranthene, measured at different temperatures. The higher temperature gas-phase experimental spectra have been taken from the NIST Chemistry WebBook data base (Linstrom & Mallard 2001), while the lower temperature spectra come from matrix isolation experiments of Hudgins & Sandford (1998a) and Hudgins & Allamandola (1999). We note that the gas-phase IR spectra do not include explicitly stated temperature values, but prior studies (e.g. Mackie et al. 2015, 2016) assumed an approximate measurement temperature of about 300 K. Fig. 6 shows the NIST spectra (grey curves), referred to as T_{high} , and the low-temperature spectra (black curves), referred to as T_{low} . Comparison of the two sets of experimental spectra measured at different temperatures reveals that thermal effects lead to notable changes in spectral characteristics, such as band broadening and peak shifts (indicated by dashed lines), which aligns with previous observations of Chen (2018).

The two pairs of experimental spectra shown in the top panels of Fig. 6a and b are compared with three theoretical spectra in the bottom panels, calculated using MLMD at 300 (red) and 50 K (blue), as well as by the anharmonic DFT method (green). Compared with the 50 K MLMD results, the temperature-induced band broadening and peak shift behaviours are better captured by the MLMD results at 300 K, with relatively low RMSE values of 0.0565 and 0.0728 for the two molecules. In contrast, the agreement with the gas-phase experimental spectra is poorer for the 50 K results.

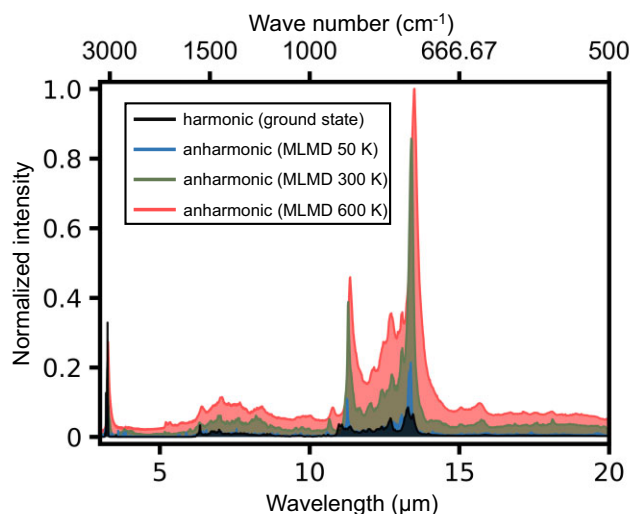


Figure 7. Absorption spectra of a mixture of 1704 neutral PAHs computed using MLMD at 50 K, 300 K, and 600 K. The three MLMD spectra were first normalized with respect to the total intensity of the 50 K spectrum, while the scaled harmonic spectrum from PAHdb was normalized to its own total intensity.

Despite the growing need to incorporate anharmonicity and temperature effects into astronomical spectroscopic analyses, driven by high-resolution IR observations in the *JWST* era (Peeters et al. 2021), only a limited number of theoretical anharmonic spectra for PAHs, likely fewer than 100, are available in the literature. Typically, these studies focus on just 2–5 PAH species, most of which contain fewer than 20 carbon atoms (Mackie et al. 2021, 2022; Esposito et al. 2024c, d). This narrow scope makes it difficult to draw general conclusions, especially given the vast diversity of PAH chemical structures. Such limitations hinder the accurate interpretation of AIBs and are mainly due to the high computational cost of traditional QCCs. To address this gap, we have computed the anharmonic spectra at 50, 300, and 600 K for 1704 molecules from version 3.20 of PAHdb, which includes neutral PAHs as large as 216 carbon atoms. This data set is open to the public as described in the section of Data Availability.

This data set is expected to be a valuable resource for the spectral decomposition of AIBs and for data-driven studies investigating the precise structure-spectral relationships of PAHs. In addition, it will aid in the exploration of temperature effects. For instance, Fig. 7 displays the anharmonic IR spectra of a simple mixture of the 1704 neutral PAHs at 50 K, 300 K, and 600 K. These spectra represent the sum of individual PAH spectra and are compared to the ground-state harmonic spectra from PAHdb (with scaling factors). Notably, the IR signals are significantly enhanced at high temperatures, exhibiting stronger peaks and an elevated plateau. A redshift in peak positions is also observed at high temperatures as a general trend. Furthermore, some emission bands that are weak at lower temperatures become significantly stronger at 300 and 600 K. For example, a distinct band emerges in the 3.5–6 μm region, absent in most harmonic spectra but observed in AIBs (Jourdain de Muizon et al. 1986; Sloan et al. 1997). This feature may originate from the redshift of the C–H stretching band in superhydrogenated PAHs due to anharmonic effects (Sandford, Bernstein & Materese 2013; Mackie et al. 2018a; Yang, Li & Glaser 2020), with additional redshift and intensity enhancement likely driven by temperature effects.

4 CONCLUSIONS

We have demonstrated that, when applied to the prediction of the IR spectra of PAHs, MLMD provides predictive accuracy comparable to that of traditional quantum chemical methods but at a fraction of the computational cost. With a scaling law of approximately $N_C^{0.7}$, MLMD is particularly suitable for large PAHs, which are key contributors to AIBs. In contrast, traditional quantum chemical anharmonic methods scale much more steeply, approximately $N_C^{4.1}$. Furthermore, we have shown that MLMD is an ideal method for studying temperature effects in molecular IR spectra, which might be critical for understanding astronomical observations. By calculating the anharmonic spectra of 1704 neutral PAHs from PAHdb at 50, 300, and 600 K, with molecular sizes containing up to 216 carbon atoms, we highlight that the efficiency of MLMD enables the creation of extensive molecular anharmonic spectral data sets. These data sets could be of instrumental importance for the advancement of AI-assisted astronomical analyses of observational IR spectra in the future.

Despite the impressive efficiency of MLMD, several limitations persist. First, the current model was trained exclusively on neutral PAHs with natural chemical elements and does not account for charged molecules or isotopologues, both of which may play a significant role in astronomical contexts. This limitation stems from the scarcity of data for the IR spectra, as well as challenges in incorporating charge state and isotope information into molecular descriptors. Second, because of the data-driven nature of the approach, the model may exhibit increased uncertainty when predicting spectra for molecules that deviate substantially from those in the training data set. While MLMD captures a broad spectrum of anharmonic effects through classical sampling of the PES, it does not account for quantum-specific phenomena such as Fermi resonance, which arise from interactions between quantized vibrational states and require a fully quantum mechanical treatment. Third, the training data of the current model were mainly sourced from molecular conformations at 300 K, which are in principle not sufficient for reliably capturing IR spectral features at very high temperatures (e.g. >600 K). We are actively working to address these challenges, with the goal of further enhancing MLMD’s capabilities for IR spectrum computation.

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DATA AVAILABILITY

The source code and the MLMD model to calculate the anharmonic spectra of PAHs using MLMD are freely available at DOI: 10.5281/zenodo.14998197. The model is ready for use without the need for further ML training. It will be continuously updated to improve predictive performance. This repository also includes the spectral data for the 1704 theoretically calculated and 49 experimentally-tested PAHs. The Supplementary Information is also included in this repository.

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SUPPORTING INFORMATION

Supplementary data are available at *MNRAS* online.

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